Chapter ten: Email server installation and setup

10.1 How to Setup a Complete Mail Server (Postfix) using 'SquirrelMail' (Webmail) on Ubuntu/Debian

Creating a mail server on Linux powered machines can be one of the most essential things that every system administrator needs to do while configuring his servers for the first time, if you don't know what it means; it's simple, if you have a website like "**example.com**", you can create an email account like "**username@example.com**" to use it to send / receive emails easily instead of using services like **Hotmail**, **Gmail**, **Yahoo Mail**, etc.

Step 1: Installing Apache2 and PHP7

1. In order to create a running mail server using "**SquirrelMail**", we'll have to install both **Apache2** & **PHP7** packages first, to do so, run.

```
$ sudo apt-get update -y
$ sudo apt-get upgrade -y
$ sudo apt-get install apache2 php7
```

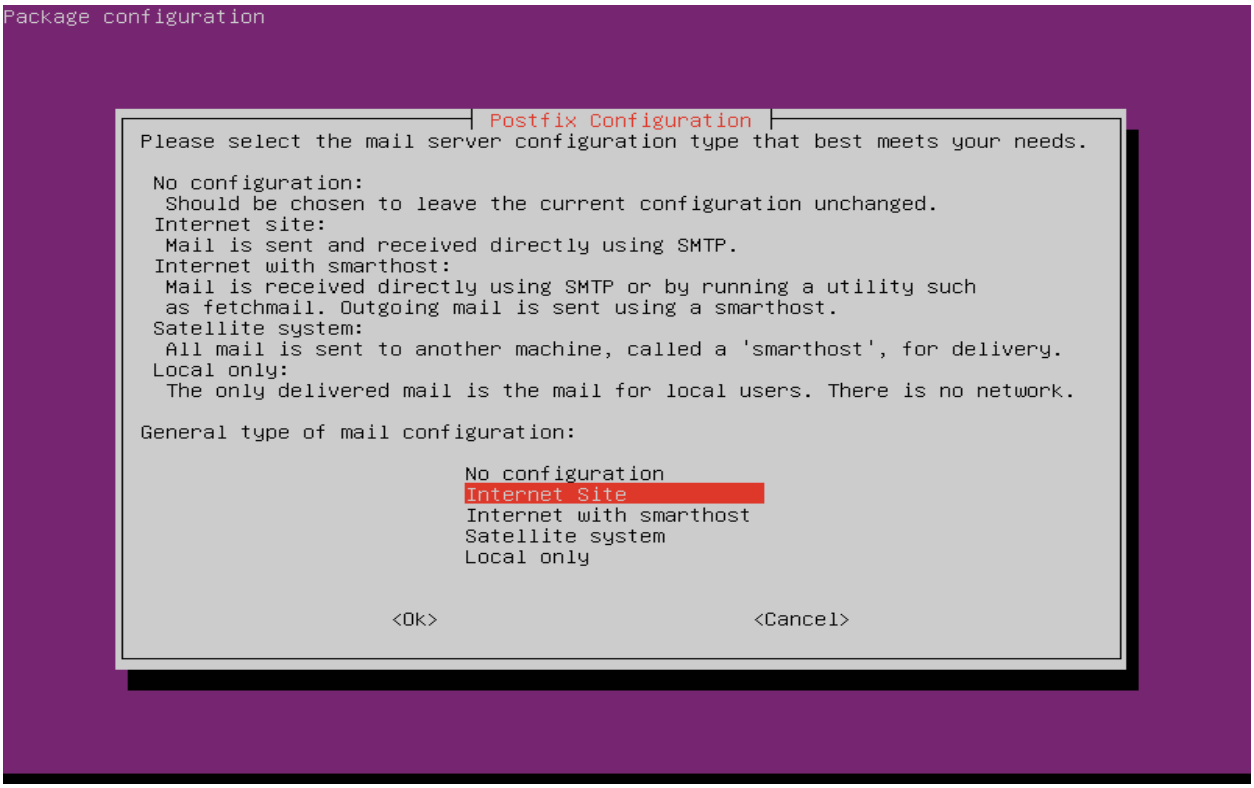
```
tecmint@ubuntu18:~$
tecmint@ubuntu18:~$ sudo apt-get install apache2 php7
Reading package lists... Done
Building dependency tree
Reading state information... Done
E: Unable to locate package php7
tecmint@ubuntu18:~$ sudo apt-get install apache2 php
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data libapache2-mod-php7.2 libaprutil1-dbd-sqlite3 libaprutil1-ldap
  liblua5.2-0 libsodium23 php-common php7.2 php7.2-cli php7.2-common php7.2-json php7.2-opcache
  php7.2-readline
Suggested packages:
  www-browser apache2-doc apache2-suexec-pristine | apache2-suexec-custom php-pear
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data libapache2-mod-php7.2 libaprutil1-dbd-sqlite3 libaprutil1-ldap
  liblua5.2-0 libsodium23 php php-common php7.2 php7.2-cli php7.2-common php7.2-json
  php7.2-opcache php7.2-readline
0 upgraded, 16 newly installed, 0 to remove and 6 not upgraded.
Need to get 5,460 kB of archives.
After this operation, 23.6 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

Step 2: Installing Postfix Mail Server

2. Postfix is a mail transfer agent (MTA) which is the responsible software for delivering & receiving emails, it's essential in order to create a complete mail server.
To install it on Ubuntu/Debian or even Mint, run:

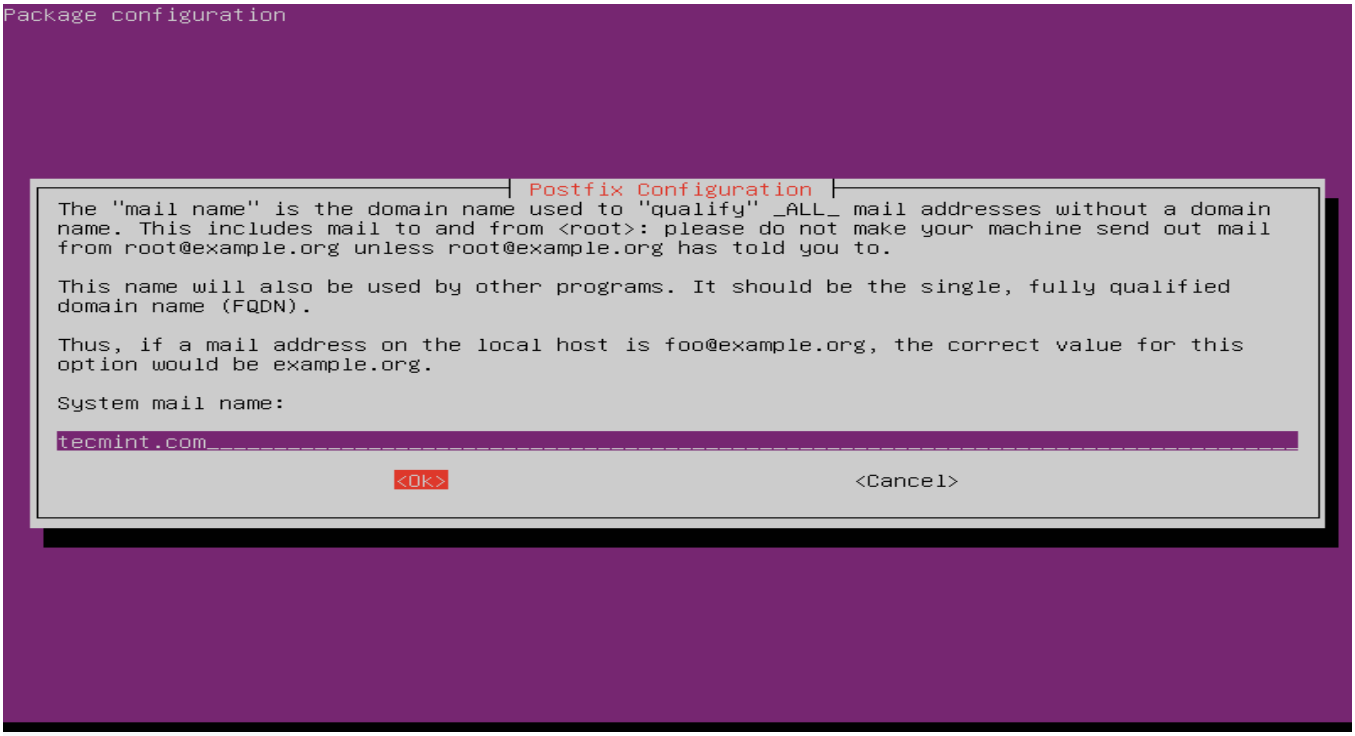
```
$ sudo apt-get install postfix
```

During installation, you will be asked to choose the type of mail configuration, choose "**Internet Site**".



Install Postfix in Ubuntu

3. Now enter the fully qualified domain name that you want to use for send and receive mails.



Set Postfix Mail Domain

1. Once the FQDN set, you've restart the Postfix mail server using.

```
$ sudo service postfix restart
```

```
tecmint@ubuntu18:~$
tecmint@ubuntu18:~$ sudo service postfix restart
tecmint@ubuntu18:~$ sudo service postfix status
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/lib/systemd/system/postfix.service; enabled; vendor preset: enabled)
   Active: active (exited) since Sat 2019-07-06 05:44:04 UTC; 4s ago
  Process: 2420 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
 Main PID: 2420 (code=exited, status=0/SUCCESS)

Jul 06 05:44:04 ubuntu18.04 systemd[1]: Starting Postfix Mail Transport Agent...
Jul 06 05:44:04 ubuntu18.04 systemd[1]: Started Postfix Mail Transport Agent.
tecmint@ubuntu18:~$ _
```

Start Postfix Mail Server in Ubuntu

Step 3: Installing Dovecot

5. Dovecot is a mail delivery agent (**MDA**), it delivers the emails from/to the mail server, to install it, run the following command.

```
$ sudo apt-get install dovecot-imapd dovecot-pop3d
```

```
tecmint@ubuntu18:~$
tecmint@ubuntu18:~$ sudo apt-get install dovecot-imapd dovecot-pop3d
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  dovecot-core libexttextcat-2.0-0 libexttextcat-data
Suggested packages:
  dovecot-gssapi dovecot-ldap dovecot-lmtpd dovecot-managesieved dovecot-mysql dovecot-pgsql
  dovecot-sieve dovecot-solr dovecot-sqlite ntp
The following NEW packages will be installed:
  dovecot-core dovecot-imapd dovecot-pop3d libexttextcat-2.0-0 libexttextcat-data
0 upgraded, 5 newly installed, 0 to remove and 6 not upgraded.
Need to get 3,099 kB of archives.
After this operation, 9,918 kB of additional disk space will be used.
Do you want to continue? [Y/n] y_
```

Install Dovecot in Ubuntu

6. Next, restart Dovecot service using the following command.

```
$ sudo service dovecot restart
```

```
tecmint@ubuntu18:~$
tecmint@ubuntu18:~$ sudo service dovecot restart
tecmint@ubuntu18:~$ sudo service dovecot status
● dovecot.service - Dovecot IMAP/POP3 email server
   Loaded: loaded (/lib/systemd/system/dovecot.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2019-07-06 05:49:09 UTC; 3s ago
     Docs: man:dovecot(1)
           http://wiki2.dovecot.org/
  Process: 6328 ExecStop=/usr/bin/doveadm stop (code=exited, status=0/SUCCESS)
 Main PID: 6331 (dovecot)
    Tasks: 4 (limit: 1711)
   CGroup: /system.slice/dovecot.service
           ├─6331 /usr/sbin/dovecot -F
           ├─6340 dovecot/anvil
           ├─6341 dovecot/log
           └─6343 dovecot/config

Jul 06 05:49:09 ubuntu18.04 systemd[1]: Started Dovecot IMAP/POP3 email server.
Jul 06 05:49:09 ubuntu18.04 dovecot[6331]: master: Dovecot v2.2.33.2 (d6601f4ec) starting up for ima
lines 1-16/16 (END)_
```

Start Dovecot in Ubuntu

Step 4: Installing Squirrel Mail in Ubuntu

7. SquirrelMail is the email server that you'll be using to manage emails on your server, it has a simple web interface to do the job, it can be customized by installing more modules & themes.

```
$ wget
https://sourceforge.net/projects/squirrelmail/files/stable/
1.4.22/squirrelmail-webmail-1.4.22.zip

$ unzip squirrelmail-webmail-1.4.22.zip

$ sudo mv squirrelmail-webmail-1.4.22 /var/www/html/

$ sudo chown -R www-data:www-data
/var/www/html/squirrelmail-webmail-1.4.22/

$ sudo chmod 755 -R /var/www/html/squirrelmail-webmail-
1.4.22/

$ sudo mv /var/www/html/squirrelmail-webmail-1.4.22/
/var/www/html/squirrelmail
```

8. After the installation, you will have to run this command in order to configure SquirrelMail.

```
$ sudo perl /var/www/html/squirrelmail/config/conf.pl
```

```
SquirrelMail Configuration : Read: config_default.php (1.4.0)
---------------------------------------------------------
Main Menu --
1.  Organization Preferences
2.  Server Settings
3.  Folder Defaults
4.  General Options
5.  Themes
6.  Address Books
7.  Message of the Day (MOTD)
8.  Plugins
9.  Database
10. Languages

D.  Set pre-defined settings for specific IMAP servers

C   Turn color on
S   Save data
Q   Quit

Command >>
```

Configure Squirrelmail in Ubuntu

09. Next, enter "**2**" in order to edit the server settings, and you will be prompted to it.

```
SquirrelMail Configuration : Read: config_default.php (1.4.0)
---------------------------------------------------------
Server Settings

General
-------
1.  Domain                  : example.com
2.  Invert Time             : false
3.  Sendmail or SMTP        : SMTP

A.  Update IMAP Settings    : localhost:143 (other)
B.  Update SMTP Settings    : localhost:25

R   Return to Main Menu
C   Turn color on
S   Save data
Q   Quit

Command >>
```

Squirrelmail Server Settings

10. Now enter "**1**" in order to change the domain name and write up your domain (e.g: **tecmint.com**).

```
SquirrelMail Configuration : Read: config_default.php (1.4.0)
---------------------------------------------------------
Server Settings

General
-------
1.  Domain                 : tecmint.com
2.  Invert Time            : false
3.  Sendmail or SMTP       : SMTP

A.  Update IMAP Settings   : localhost:143 (other)
B.  Update SMTP Settings   : localhost:25

R   Return to Main Menu
C   Turn color on
S   Save data
Q   Quit

Command >> 1

The domain name is the suffix at the end of all email addresses.  If
for example, your email address is jdoe@example.com, then your domain
would be example.com.

[tecmint.com]: tecmint.com
```

Set Mail Domain

11. Go back to the main menu by writing "**R**" and hitting the enter key, write "**4**" in order to configure the general options.

You see "**Allow server-side sorting**"? Enter "**11**" and change it from "**false**" to "**true**" by entering "**y**". Now hit the **Enter** key, and enter the "**S"** key in order to save the configuration file.

```
SquirrelMail Configuration : Read: config_default.php (1.4.0)
---------------------------------------------------------
General Options
1.  Data Directory                : /var/local/squirrelmail/data/
2.  Attachment Directory          : /var/local/squirrelmail/attach/
3.  Directory Hash Level          : 0
4.  Default Left Size             : 150
5.  Usernames in Lowercase        : false
6.  Allow use of priority         : true
7.  Hide SM attributions          : false
8.  Allow use of receipts         : true
9.  Allow editing of identity     : true
    Allow editing of name         : true
    Remove username from header   : false
10. Allow server thread sort      : false
11. Allow server-side sorting     : false
12. Allow server charset search   : true
13. Enable UID support            : true
14. PHP session name              : SQMSESSID
15. Location base                 :
16. Only secure cookies if poss.  : true
17. Disable secure forms          : false
18. Page referal requirement      :

R   Return to Main Menu
C   Turn color on
S   Save data
Q   Quit

Command >> 11

This option allows you to choose if SM uses server-side sorting
Your IMAP server must support the SORT  command for this to work

Allow server-side sorting? (y/n) [n]: y_
```

Configure General Mail Options

12. You can now access the mail server by going to **domain.com/squirrelmail**.

Access Squirrelmail

Step 5: Creating Mail Users

13. In order to start using squirrelmail webmail, you'll have to create a new user, to do so, run.

```
$ sudo useradd myusername
```

Replace "**myusername**" with the user name you want, create a password for the new user by running.

```
$ sudo passwd myusername
```

Replace "**myusername**" with the user name you want, create a password for the new user by running.

```
$ sudo passwd myusername
```

14. Create a home folder for the user in **/var/www/html/myusername** and make it default home directory.

```
$ sudo mkdir -p /var/www/html/myusername
$ usermod -m -d /var/www/html/myusername myusername
```

15. Now go back to the login page and enter the user name and the password of newly created user.

Access Squirrelmail

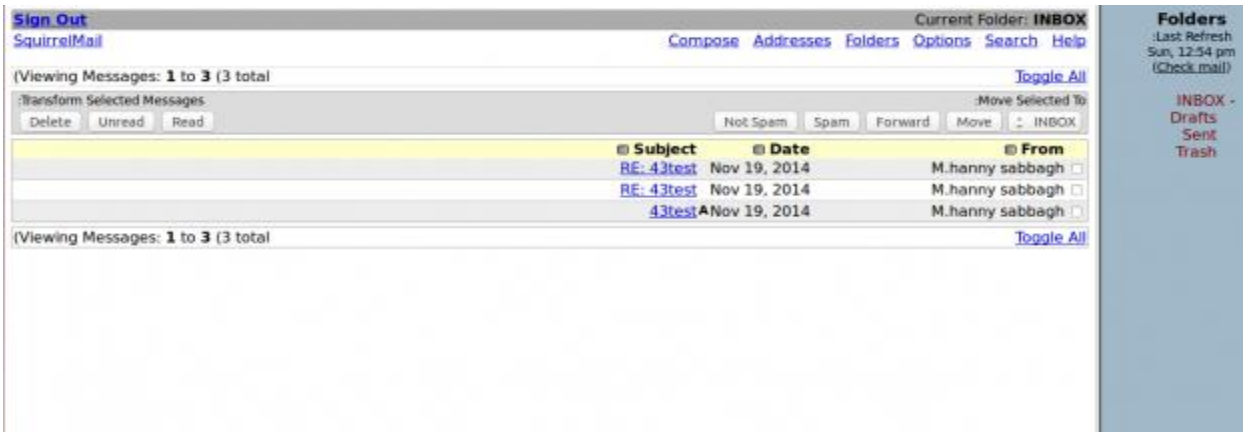
You will be surprise to see the following error message.

10.2Login to Squirrelmail

This is just a problem in the permissions, you have to give the user "**myusername**" the complete permissions on its home folder.

```
$ sudo chown -R myusername:myusername /var/www/html/myusername
```

16. Once permission set, you should able to login into squirrelmail.



Squirrelmail Mail Interface

You can try to send email from it, or you can try to receive emails by sending it to "**myusername@example.com**", don't forget to replace "**myusername**" with the user name you created.If you faced any other error.. Just check the "**/var/log/mail.err**" file, all the error message will be stored there, you won't lose your way :)